
Xavier Vicent Navarro

Robotics & Autonomous Systems Engineer

201 NE 2nd Ave, Dania Beach, 33004, FL
906-287-9897
xavitenis27@gmail.com

Professional Summary

Marine robotics and autonomous systems engineer pursuing a Ph.D. in Ocean Engineering, with 5+ years of experience developing autonomy, perception, navigation, control, and multi-vehicle coordination systems for unmanned surface and aerial vehicles. Experienced in ROS 2 software development, sensor fusion, computer vision, simulation, hardware integration, and maritime field testing. Published research includes USV control, autonomous navigation, multi-vehicle collision avoidance, station-keeping, and UAV launch and recovery from moving maritime platforms.

Education

Florida Atlantic University

Ph.D. in Ocean Engineering (Robotics & AI Focus), 2025–2027

Dissertation: *Decentralized Cooperative Navigation of USV–UAV Teams in the Maritime Domain*
Advisor: Dr. Manhar R. Dhanak

Florida Atlantic University

M.S. in Ocean Engineering (Robotics & AI), 2024–2025

Thesis: *PID Auto-Tuning for USVs: A Multi-Evolutionary Optimization AI Approach*
Advisor: Dr. Manhar R. Dhanak
Minor: Artificial Intelligence

Lake Superior State University

B.S. in Robotics Engineering, 2020–2023

Senior Design: *Launch and Recover a UAV from a USV with ROS and PX4*
Minors: Mathematics and Electrical Engineering

Research and Technical Experience

Graduate Research Assistant

Florida Atlantic University, Dania Beach, FL | Jan 2024 – Present

- Conduct doctoral research on decentralized navigation, perception, advanced control, and coordination for USV–UAV teams in maritime environments.
- Develop ROS 2 software for path planning, trajectory tracking, COLREGs-based collision avoidance, and multi-vehicle coordination.
- Design and implement linear and nonlinear control methods for path following, station-keeping, heading and speed regulation, and disturbance rejection.
- Develop perception, localization, mapping, and sensor-fusion pipelines using LiDAR, stereo vision, IMU, RTK-GPS, and deep learning.
- Validate autonomy and control software through simulation, hardware-in-the-loop testing, and field experiments on physical USV and UAV platforms.

Incoming Visiting Researcher – NATO Visiting Research Programme (VRP)

NATO STO Centre for Maritime Research and Experimentation (CMRE), La Spezia, Italy | Sep 2026–Nov 2026

- Selected to conduct dissertation research on cooperative autonomy, UAV perception, and cross-domain coordination for unmanned maritime systems.
 - Develop onboard UAV autonomy, ROS 2 ground-station integration, and perception methods for coordinated multi-domain operations.
-

USV Control Engineer Intern

Lake Superior State University, Sault Ste. Marie, MI | Dec 2022 – Dec 2023

- Developed and tested ROS-based navigation and control software for unmanned surface vehicles.
- Validated motion planning and control in simulation prior to field deployment.
- Tuned and debugged autonomy systems under disturbances and sensor noise.
- Supported integration of control, sensing, and software systems on robotic platforms.

Controls Engineer Intern

Accent Controls Inc., North Kansas City, MO | Jun 2022 – Aug 2022

- Designed electrical schematics and PLC logic for industrial control systems.
- Developed diagnostic and lockout/tagout safety documentation for automated plants.

Skills**Programming and Development**

Python, C++, MATLAB, Linux, Git, Bash, PyQt

Robotics Middleware and Simulation

ROS, ROS 2, Gazebo, RViz, PX4, MAVROS, DDS

Autonomy, Guidance, Navigation, and Control

Path planning, trajectory generation, path tracking, localization, state estimation, sensor fusion, PID control, nonlinear control, station-keeping, obstacle avoidance, COLREGs-based navigation, and multi-robot coordination

Artificial Intelligence and Perception

Computer vision, YOLO, convolutional neural networks, reinforcement learning, evolutionary optimization, object detection, semantic segmentation, and camera-LiDAR fusion

Sensors and Embedded Systems

LiDAR, stereo cameras, IMUs, RTK-GPS, sonar and depth sensors, NVIDIA Jetson, Raspberry Pi, Arduino, and STM32

Robotic Platforms

Unmanned surface vehicles, unmanned aerial vehicles, remotely operated vehicles, and autonomous underwater systems

Publications

- **X. Vicent Navarro**, M.R. Dhanak, and E. I. Sarda, "PID auto-tuning for USVs: A multi-evolutionary optimization AI approach," *Ocean Engineering*, vol. 355, part 1, 2026, Art. no. 124987.
- **X. Vicent Navarro**, and M.R.Dhanak, "Decentralized COLREGs-Based Navigation for Multiple USVs Using Artificial Potential Fields", Chicago IEEE Oceans Conference 2025, Chicago, USA
- **X. Vicent Navarro**, E. I. Sarda, and M.R.Dhanak, "Fuzzy Logic-Assisted PID Control for Improved Station-Keeping of a Catamaran-Style USV", Chicago IEEE Oceans Conference 2025, Chicago, USA
- **X. Vicent Navarro**, and M.R.Dhanak, "Performance and comparison of PID-based techniques for USV navigation" Halifax IEEE Oceans Conference 2024, Halifax, Canada
- **X. Vicent Navarro**, C. Wilson, and E. I. Sarda, "Accurate Autonomous Launch and Recovery of an UAV from a Station Keeping USV" Halifax IEEE Oceans Conference 2024, Halifax, Canada
- Z. Kassuba, M. Goldenbogen, C. Jannetta, **X. Vicent Navarro**, and E. I. Sarda, "Development of a Water Quality Monitoring System on a USV", Halifax IEEE Oceans Conference 2024, Halifax, Canada
- **X. Vicent Navarro**, and M.R. Dhanak, "AI-Assisted Real-Time Vision-LiDAR Semantic Mapping of Maritime Obstacles for Autonomous USV Navigation", Monterey IEEE Oceans Conference 2026, California, Canada, in Preparation
- **X. Vicent Navarro**, and M.R. Dhanak, "AI-Assisted Dynamic Model Identification of Unmanned Surface Vehicles", *Journal of Ocean Engineering*, In Preparation
- **X. Vicent Navarro**, and M.R. Dhanak, "Dynamics-Based Artificial Potential Field Control of Unmanned Surface Vehicles", *Journal of Ocean Engineering*, In Preparation

Leadership & Extracurricular Activities

- **President**, FAU Marine Robotics Club (2024–Present); led student activities and team participation in **RoboBoat** and **RobotX** competitions.
 - **President**, LSSU RobotX Club (2022–2023); supported development and competition preparation for **RobotX** and **VRX**.
 - **Captain**, LSSU Varsity Men's Tennis Team (2021–2023); named **Second Team All-Conference** in 2022.
-